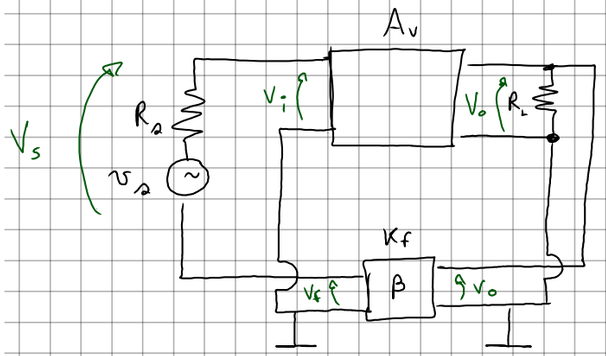
MVSV (Voltage - Series)



$V_f = V_o \cdot \beta$, $A_v = \frac{V_o}{V_i}$ (a lazo abierto)

$\rightarrow \beta = \frac{V_f}{V_o}$

$A_{Tot} = \frac{V_o}{V_s}$, (a lazo cerrado) Con K_f negativo $\rightarrow V_i = V_s - V_f$

$V_i = V_s - V_o \cdot \beta$

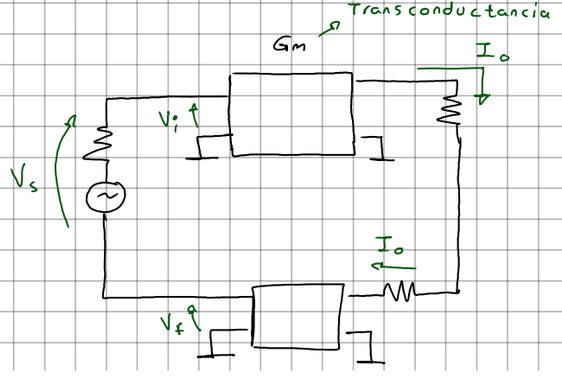
$V_o = A_v \cdot V_i = A_v (V_s - V_o \cdot \beta)$

$V_o = A_v \cdot V_s - A_v \cdot V_o \cdot \beta$

$\rightarrow V_o + A_v \cdot V_o \cdot \beta = A_v \cdot V_s$

$\rightarrow \frac{V_o}{V_s} = \frac{A_v}{1 + A_v \cdot \beta}$

MISV (Current - Series , $\frac{I}{V}$)



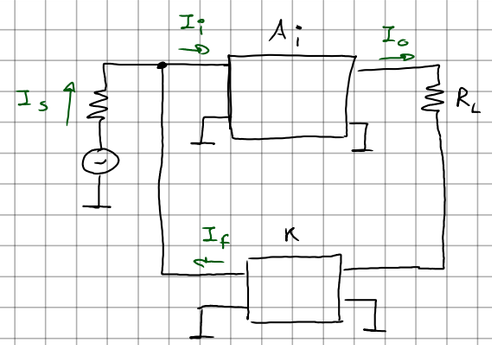
$G_m = \frac{I_o}{V_i}$, $\beta = \frac{V_f}{I_o}$

$V_i = V_s + V_f$

$G_{Tot} = \frac{I_o}{V_s} = \frac{I_o}{V_i - V_f} = \frac{I_o}{\frac{I_o}{G_m} - \beta I_o}$

$\rightarrow G_T = \frac{G_m}{1 - \beta G_m}$

MIMI (Current - Shunt)



Lazo Abierto

$A_i = \frac{I_o}{I_i}$, $\beta = \frac{I_f}{I_o}$

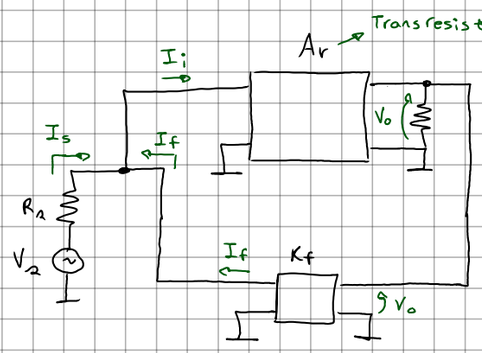
$A_{Tot} = \frac{I_o}{I_s}$, $I_s = I_i - I_f = I_i - \beta I_o = I_i - \beta \cdot A_i \cdot I_i$

$I_o = A_i \cdot I_i$

$A_T = \frac{A_i \cdot I_i}{I_i - \beta A_i I_i}$

$A_T = \frac{A_i}{1 - \beta A_i}$

MVSI (Voltage - Shunt , $\frac{V}{I}$)



$A_f = \frac{V_o}{I_i}$, $\beta = \frac{I_f}{V_o}$

$I_i = I_s + I_f$, $I_f = \beta V_o = \beta \cdot I_i \cdot A_r = \beta \cdot A_r \cdot (I_s + I_f)$

$\rightarrow V_o = A_r I_i = A_r (I_s + I_f) = A_r (I_s + \beta V_o)$

$V_o = A_r I_s + A_r \beta V_o$

$\rightarrow V_o (1 - A_r \beta) = A_r I_s$

$\rightarrow A_T = \frac{V_o}{I_s} = \frac{A_r}{1 - A_r \beta}$ ✓

$\rightarrow A_{Tot} = \frac{V_o}{I_s} = \frac{V_o}{I_i - I_f} = \frac{V_o}{\frac{V_o}{A_r} - \beta V_o} = \frac{1}{\frac{1}{A_r} - \beta} \cdot \frac{\frac{A_r}{A_r}}{\frac{A_r}{A_r}} = \frac{A_r}{1 - A_r \beta}$ ✓